(Past paper)
(2018)

C) Find the pdf of $Y = x_1 + x_2 + \cdots + x_n$ if F is a F-distributed random variable with n_1 & n_2 df then show that

$$y = \left(1 + \frac{n_1}{n_2} F\right)^{-1} \text{ has beta dist of } 1^{st} \text{ kind}$$

(solution)

$$d\rho(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}} dF \quad 0 < F < \infty$$

using

$$y = \left(1 + \frac{n_1}{n_2} F\right)^{-1}$$

$$1/y = \left(1 + \frac{n_1}{n_2} F\right)$$

$$1/y - 1 = n_1/n_2 \, F$$

$$F = \left(\frac{n_2}{n_1}\right)\left(1/y - 1\right)$$

$$dF = -\, n_2/n_1 \; 1/y^2 \, dy \qquad 1 < y < 0$$

$$d\rho(y) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \left(\left(\frac{n_2}{n_1}\right)\left(1/y - 1\right)\right)^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\left(1/y\right)^{\frac{n_1+n_2}{2}}} \cdot \left(\frac{-n_2}{n_1}\right)\left(1/y^2\right) dy$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\cancel{\frac{n_1}{2}} - \cancel{\frac{n_1}{2}} + \cancel{1} - \cancel{1}} \left(1 - y/y\right)^{\frac{n_1}{2}-1} \cdot 1/y^2 \, dy}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\left(1/y\right)^{\frac{n_1+n_2}{2}}}$$

Scanned with CamScanner

$$= \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} (1-y)^{\frac{n_1}{2}-1} \cdot y^{1-\cancel{\frac{n_1}{2}}-2+\cancel{\frac{n_1}{2}}+\frac{n_2}{2}} dy$$

$$= \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} (1-y)^{\frac{n_1}{2}-1} \cdot y^{\frac{n_2}{2}-1} dy$$

$$= \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} y^{\frac{n_2}{2}-1} (1-y)^{\frac{n_1}{2}-1} dy$$

$$f(y) = \frac{1}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} y^{\frac{n_2}{2}-1} (1-y)^{\frac{n_1}{2}-1} \qquad 0<y<1$$

---

d)- if x is a random variable with p.d.f as

$$f(x) = \frac{1}{2} \qquad -1 \le x \le 1$$

Find the p.d.f of $Y = x^2$

a) Find the mean and variance of pareto-distribution

(sol)

By defination

$$\mu_r' = E(x^r) = \int_0^\infty x^r \cdot f(x)\,dx$$

$$= \cancel{\frac{1}{\beta(\alpha,\beta)}} = \alpha k^\alpha \int_k^\infty x^r \cdot x^{-\alpha-1}$$

$$= \alpha k^\alpha \int_k^\infty x^{r-\alpha-1}\,dx$$

$$= \alpha k^\alpha \left| \frac{x^{r-\alpha}}{r-\alpha} \right|_k^\infty$$

$$= \alpha k^\alpha \left((\infty)^{r-\alpha} - k^{r-\alpha} \right)$$

$$= -\frac{\alpha k^\alpha \cdot k^r \cdot k^{-\alpha}}{r-\alpha}$$

$$\mu_r' = \frac{\alpha k^r}{r-\alpha}$$

put $r=1$

$$\mu_1' = \frac{\alpha k}{1-\alpha}$$

Mean:-

$$\mu_1 = \mu_1' - (\mu_1')^{2}$$

$$\mu_1' = 0$$

Variance:-

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_2' = \frac{\alpha k^2}{\alpha - 2}$$

$$= \frac{\alpha k^2}{\alpha - 2} - \left(\frac{\alpha k}{\alpha - 1}\right)^2$$

$$= \alpha k^2 \left[\frac{1}{\alpha - 2} - \frac{\alpha}{(\alpha - 1)^2}\right]$$

$$= \alpha k^2 \left[\frac{(\alpha - 1)^2 - (\alpha - 2)\alpha}{(\alpha - 2)(\alpha - 1)^2}\right]$$

$$= \alpha k^2 \left[\frac{\cancel{\alpha^2} + 1 - \cancel{2\alpha} - \cancel{\alpha^2} + \cancel{2\alpha}}{(\alpha - 2)(\alpha - 1)^2}\right]$$

$$\text{Variance} = \alpha k^2 / (\alpha - 2)(\alpha - 1)^2$$

b)- IF $x_1, x_2, \ldots, x_n$ are "n" independent random variable having exponential dist having same parameter 'θ' i.e $f(x) = \frac{1}{\theta} e^{-x/\theta}$

Sol:-

$$f(x) = \frac{1}{\theta} e^{-x/\theta}$$

$$F(x) = 1 - e^{-x/\theta}$$

Scanned with CamScanner

The r^{th} order statistic is

$$g(y_r) = \frac{n!}{(r-1)!\,(n-r)!}\left[F(y_r)\right]^{r-1} f(y_r)\left[1-F(y_r)\right]^{n-r}$$

→ put $r=1$ and $n=n$

$$g(y_1) = \frac{n!}{(1-1)!\,(n-1)!}\left[F(y_1)\right]^{1-1} f(y_1)\left[1-F(y_1)\right]^{n-1}$$

$$= \frac{n\cancel{(n-1)!}}{\cancel{(n-1)!}} f(y_1)\left[1-F(y_1)\right]^{n-1}$$

$$= n\left(1/\theta\, e^{-y_1/\theta}\right)\left[1-(1-e^{-y_1/\theta})\right]^{n-1}$$

$$= \frac{n}{\theta}\left(e^{-y_1/\theta}\right)\left[\cancel{1}-\cancel{1}+e^{-y_1/\theta}\right]^{n-1}$$

$$= \frac{n}{\theta}\left(e^{-y_1/\theta}\right)\left(e^{-y_1/\theta}\right)^{n-1}$$

$$= \frac{n}{\theta}\left(e^{-y_1/\theta-ny_1/\theta+y_1/\theta}\right)$$

$$g(y_1) = n/\theta\left(e^{-ny_1/\theta}\right) \qquad 0<y_1<\infty$$

To obtain the distribution of y_n, put $r=n$, $n=n$

$$g(y_n) = \frac{n!}{(n-1)!\,(n-n)!}\left[F(y_n)\right]^{n-1} f(y_n)\left[1-F(y_n)\right]^{n-n}$$

$$= \frac{n\cancel{(n-1)!}}{\cancel{(n-1)!}} f(y_n)\left[F(y_n)\right]^{n-1}$$

$$= n\left(1/\theta\, e^{-y_n/\theta}\right)\left[1-e^{-y_n/\theta}\right]^{n-1}$$

$$g(y_n) = n/\theta\left(e^{-y_n/\theta}\right)\left[1-e^{-y_n/\theta}\right]^{n-1} \qquad 0<y_n<\infty$$

Scanned with CamScanner

Q:05:-

Let x_1 and x_2 are two independent standard normal variables. Then obtain the dist of $z = \frac{x_1}{x_2}$

-(Sol)-

The p.d.f of x_1 and x_2

$$f(x_1) = \frac{1}{\sqrt{2\pi}} e^{-1/2 x_1^2} \qquad 0 < x_1 < \infty$$

$$f(x_2) = \frac{1}{\sqrt{2\pi}} e^{-1/2 x_2^2} \qquad 0 < x_2 < \infty$$

The joint distribution of x_1 and x_2

$$dF(x_1, x_2) = \frac{1}{2\pi} e^{-1/2(x_1^2 + x_2^2)} \cdot dx_1 \, dx_2$$

As $z = x_1/x_2$

$$x_2 = y$$

$$z = x_1/y$$

$$x_1 = yz$$

$$\frac{\partial x_1}{\partial y} = z \, , \quad \partial x_1/\partial z = y$$

$$\frac{\partial x_2}{\partial y} = 1 \, , \quad \partial x_2/\partial z = 1$$

$$J = \begin{vmatrix} z & y \\ 1 & 0 \end{vmatrix}$$

$$|J| = |-y|$$

PUACP

Scanned with CamScanner

$$df(y,x) = \frac{1}{2\pi} e^{-1/2(y^2x^2+y^2)} \cdot |-y| \, dy \, dx$$

$$= \frac{1}{2\pi} e^{-1/2 y^2(x^2+1)} \cdot y \, dy \, dx$$

→ integrating w.r.t "y"

$$= \frac{1}{2\pi} dx \int_{-\infty}^{\infty} e^{-1/2 y^2(x^2+1)} y \, dy$$

→ put to even function

$$= \frac{2}{2\pi} dx \int_{0}^{\infty} e^{-1/2 y^2(x^2+1)} y \, dy$$

$$\because y^2 \frac{(x^2+1)}{2} = t$$

$$y^2 = \frac{2t}{x^2+1}$$

$$2y \, dy = \frac{2}{x^2+1} dt$$

$$y \, dy = \frac{dt}{1+x^2}$$

$$= \frac{1}{\pi} dx \int_0^{\infty} e^{-t} \cdot \frac{dt}{1+x^2}$$

$$= \frac{1}{\pi(1+x^2)} dx \int_0^{\infty} e^{-t} \cdot t^{1-1} \, dt$$

$$= \frac{1}{\pi(1+x^2)} dx \, \Gamma 1$$

$$f(x) = \frac{1}{\pi(1+x^2)}$$

Scanned with CamScanner

Q # 03:-

Drive χ^2- distribution

(Sol)

let us take the two variables x_1 and x_2

$$f(x_1) = \frac{1}{\sqrt{2\pi}} e^{-1/2 \cdot x_1^2} \quad dx \quad 0 < x < \infty$$

$$f(x_2) = \frac{1}{\sqrt{2\pi}} e^{-1/2 \cdot x_2^2} \quad 0 < x < \infty$$

The joint-dist of x_1 and x_2

$$dF(x_1, x_2) = \frac{1}{2\pi} e^{-1/2(x_1^2 + x_2^2)} \cdot dx_1, dx_2$$

$$x_1 = r_2 \cos\theta_1, \quad \partial x_1/\partial r_2 = \cos\theta_1, \quad \partial x_1/\partial\theta = -r_2 \sin\theta$$

$$x_2 = r_2 \sin\theta, \quad \partial x_2/\partial r_2 = \sin\theta_1, \quad \partial x_2/\partial\theta = r_2 \cos\theta$$

$$|J| = \begin{vmatrix} \cos\theta_1 & -r_2 \sin\theta_1 \\ \sin\theta_1 & r_2 \cos\theta_1 \end{vmatrix}$$

$$= |r_2 \cos^2\theta_1 + r_2 \sin^2\theta_1|$$

$$= r_2 (\cos^2\theta_1 + \sin^2\theta_1)$$

$$|J| = r_2$$

$$dF(r_2, \theta) = c^2 e^{-1/2 r_2^2} \cdot r_2 \, d\theta_1 \, dr_2$$

$$= c^2 e^{-1/2 r_2^2} r_2 \, dr_2 \int_0^{2\pi} d\theta_1$$

$$= c^2 \cdot e^{-1/2 r_2^2} \cdot r_2 \, dr_2 \, |\theta|_0^{2\pi}$$

$$= c^2 \cdot e^{-1/2 r_2^2} r_2 \, dr_2 \, (2\pi)$$

$$= 2\pi c^2 \cdot e^{-1/2 r_2^2} \cdot r_2 \, dr_2$$

Scanned with CamScanner

$$f(x_3) = \frac{1}{\sqrt{2\pi}} e^{-1/2 x_3^2}$$

The joint distribution of r_2 and x_3

$$dF(r_2, x_3) = 2\pi c^3 e^{-1/2(r_2^2 + x_3^2)} \cdot r_2 \, dr_2 \, dx_3$$

$r_2 = r_3 \cos\theta_2$, $\partial r_2/\partial r_3 = \cos\theta_2$, $\partial r_3/\partial\theta_2 = -r_3\sin\theta_2$

$x_3 = r_3 \cos\sin\theta_2$, $\partial x_3/\partial r_3 = \sin\theta_2$, $\partial x_3/\partial\theta_2 = r_3\cos\theta_2$

$$|J| = \begin{vmatrix} \cos\theta_2 & -r_3\sin\theta_2 \\ \sin\theta_2 & r_3\cos\theta_2 \end{vmatrix}$$

$$= |r_3 \cos^2\theta_2 + r_2 \sin^2\theta_2|$$

$$= r_3 \,(1)$$

$$|J| = r_3$$

$$= 2\pi c^3 e^{-1/2 r_3^2} \cdot (r_3\cos\theta_2) r_3 \, dr_3 \, d\theta_2$$

$$= 2\pi c^3 e^{-1/2 r_3^2} r_3^2 \, dr_3 \int_{-\pi/2}^{\pi/2} (\cos\theta_2) d\theta_2$$

$$= 2\pi c^3 e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3 \left| \sin\theta_2 \right|_{-\pi/2}^{\pi/2}$$

$$= 2\pi c^3 e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3 \, |1 - (-1)|$$

$$= 2 \cdot 2\pi c^3 e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3$$

$$= 2 \cdot 2\pi^{3/2 - 1/2} c^3 \cdot e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3$$

$$= \frac{2\pi^{3/2}}{\sqrt{\pi}/2} c^3 \cdot e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3$$

$$= 2 \cdot \frac{\pi^{3/2}}{\Gamma 3/2} c^3 \cdot e^{-1/2 r_3^2} \cdot r_3^2 \, dr_3$$

Scanned with CamScanner

As

$$r_3^2 = x_1^2 + x_2^2 + x_3^2 \Rightarrow \sum_{i=1}^{3} x_i^2$$

let

$$r_m^2 = \sum_{i=1}^{m} x_i^2$$

so, the distribution of r_m

$$dF(r_m) = \frac{2\pi^{m/2}}{\Gamma m/2} \cdot c^{m+1} \cdot e^{-1/2\, r_m^2} \cdot r_m^{m-1}\, dr_m$$

$\rightarrow$ so, the dist of r_m and x_{m+1}

$$dF(r_m, x_{m+1}) = \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2(r_m^2 + x_{m+1}^2)} \cdot r_m^{m-1}\, dr_m\, dx_{m+1}$$

$$r_m = r_{m+1} \cos\theta_m$$

$$x_{m+1} = r_{m+1} \sin\theta_m$$

$$|J| = (r_{m+1})$$

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2(r_{m+1}^2)} \cdot (r_{m+1}\cos\theta_m)^{m-1} \cdot r_{m+1} \cdot dr_{m+1}\, d\theta_m$$

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2\, r_{m+1}^2} \cdot r_{m+1}^{m} (\cos\theta_m)^{m-1}\, dr_{m+1}\, d\theta_m$$

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2\, r_{m+1}^2}\, r_{m+1}^{m}\, dr_{m+1} \int_{-\pi/2}^{\pi/2} (\cos\theta_m)^{m-1}\, d\theta_m$$

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} \cdot e^{-1/2\, r_{m+1}^2} \cdot r_{m+1}^{m}\, dr_{m+1} \int_{-\pi/2}^{\pi/2} (\cos\theta)^{2(\frac{m}{2})-1} \cdot (\sin\theta_m)^{2(\frac{1}{2})-1}$$

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2\, r_{m+1}^2} \cdot r_{m+1}^{m}\, dr_{m+1}\, \beta\left(\frac{m}{2}, \frac{1}{2}\right)$$

Scanned with CamScanner

$$= \frac{2\pi^{m/2}}{\Gamma m/2} c^{m+1} e^{-1/2 r_{m+1}^2} \cdot r_{m+1}^{m} \, dr_{m+1} \cdot \frac{\Gamma \frac{m}{2} \cdot \Gamma \frac{1}{2}}{\Gamma \frac{m+1}{2}}$$

Here

$$r_{m+1}^2 = \sum_{i=1}^{m+1} x_i^2$$

$$r_{m+1}^2 = x^2$$

$$x = r_{m+1}$$

$$dx = dr_{m+1} \quad , \text{also} \quad n = m+1$$

$$n-1 = m$$

$$= \frac{2\pi^{n/2}}{\Gamma n/2} c^{n} \cdot e^{-1/2 x^2} \cdot (x^2)^{n-1} dx$$

→ To find the dist of x^2

$$x = (x^2)^{1/2 - 1} \, dx^2$$

$$dF(x^2) = \frac{1}{2} (x^2)^{1/2-1} dx^2$$

$$= \frac{2\pi^{n/2}}{\Gamma \frac{n}{2}} \frac{1}{(2\pi)^{n/2}} \cdot e^{-1/2 x^2} \cdot \left(\frac{1}{2} (x^2)^{-1/2}\right) dx^2$$

$$= \frac{1}{2^{n/2} \Gamma \frac{n}{2}} e^{-1/2 x^2} \cdot (x^2)^{n/2 - 1} \cdot dx^2$$

$$0 < x^2 < \infty$$

Scanned with CamScanner

(Probability)

(Past Paper 2019)

Q#2

i- Show that the square of t-variate with n-degree of freedom is distributed as F=1 and n degree of freedom

(Solution)-

As we know that

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{(F)^{\frac{n_2}{2}-1}}{\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}} dF$$

$0 < F < \infty$

For $n_1 = 1$ and $n_2 = n$

$F = t^2$ | $F=0$, $t = 0$

$dF = 2t\,dt$ | $F=\infty$, $t=\infty$

$$dp(t) = \frac{(1/n)^{1/2}}{\beta\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{(t^2)^{1/2-1}}{\left(1+\frac{1}{n}t^2\right)^{\frac{1+n}{2}}} 2t\,dt$$

$$= \frac{2t\,(t^2)^{-1/2}}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt \qquad 0 < t < \infty$$

$$= \frac{2}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} dt \qquad 0 < t < \infty$$

it is an even function

$$f(t) = \frac{1}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{n+1}{2}}} \qquad 0 < t < \infty$$

Scanned with CamScanner

ii- let $Y_1 < Y_2 < Y_3 < Y_4$ denotes the order statistics of a random sample of size 4 from dist having p.d.f

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & , \text{otherwise} \end{cases}$$

Find the dist of Y_3

(solution)

Given that

$$f(x) = 2x \qquad 0 < x < 1$$

$$F(x) = 2\int_0^x x\,dx$$

$$= \cancel{2} \left| \frac{x^2}{\cancel{2}} \right|_0^x$$

$$= x^2 - 0$$

$$F(x) = x^2$$

The rth order statistic is

$$g(y_r) = \frac{n!}{(r-1)!(n-r)!} [F(y_r)]^{r-1} f(y_r) [1 - F(y_r)]$$

Put $r = 3$ and $n = 4$ in eq

$$g(y_3) = \frac{4!}{(3-1)!(4-3)!} [F(y_3)]^{3-1} f(y_3) [1 - F(y_3)$$

$$= \frac{4!}{2!\,1!} (y_3^2)^2 \, 2y_3 \, [1 - y_3^2)^1$$

$$= 12 \times 2 \, (y_3^{4+1}) (1 - y_3^2)$$

$$g(y_3) = 24 \, (y_3^5 - y_3^7)$$

Find the mean and variance of pareto distribution

-(Sol)-

By defination

$$E(x) = \int_k^\infty x \cdot g(x)\,dx$$

$$= \int_k^\infty x \cdot \frac{\alpha k^\alpha}{x^{\alpha+1}}\,dx$$

$$= \alpha k^\alpha \int_k^\infty x \cdot x^{-\alpha-1}\,dx$$

$$= \alpha k^\alpha \int_k^\infty x^{\cancel{1}-\alpha-\cancel{1}}\,dx$$

$$= \alpha k^\alpha \int_k^\infty x^{-\alpha}\,dx$$

$$= \cancel{\alpha} k^\alpha \left| \frac{x^{-\alpha}}{-\cancel{\alpha}} \right|_k^\infty$$

$$= -k^\alpha \left((\infty)^{-\alpha} - k^{-\alpha} \right)$$

$$= -\cancel{k}^\alpha + k^{-\alpha}$$

mean $= E(x) = 0$

Scanned with CamScanner

iv- If the random variable X has normal-distribution with mean m and variance σ^2, obtain the distribution of $Y = aX + b$

-(solution)-

$$Y = aX + b$$

i- Mean

$$E(Y) = aE(X) + b$$

$$y = a\mu + b$$

ii- Variance

$$V(Y) = a^2 V(X)$$

$$= a^2 \sigma^2$$

iii)- let x and y are independent random variables with joint p.d.f

$f(x,y) = x + y \quad 0<x<1, 0<y<1$

$f_1 = 0$ elsewhere

Find $E(XY)$ and $E(X+Y)$

-(solution)-

$$E(x,y) = \int_0^1\int_0^1 x,y\,(x+y)\,dx\,dy$$

$$= \int_0^1 y\left\{\int_0^1 x^2\,dx + \int_0^1 xy\,dx\right\}dy$$

$$= \int_0^1\left\{y\left[\left|\frac{x^3}{3}\right|_0^1 + y\left|\frac{x^2}{2}\right|_0^1\right]\right\}dy$$

$$= \int_0^1 y(1/3) + y(1/2)\,dy$$

$$= \frac{1}{3}\int_0^1 y\,dy + \frac{1}{2}\int_0^1 y^2\,dy$$

$$= \frac{1}{3}\left| \frac{y^2}{2} \right|_0^1 + \frac{1}{2}\left| \frac{y^3}{3} \right|_0^1$$

$$= \frac{1}{6}\left| y^2 \right|_0^1 + \frac{1}{6}\left| y^3 \right|_0^1$$

$$= 1/6(1) + 1/6(1)$$

$$E(x, y) = 2/6 \Rightarrow 1/3$$

$$f_x(y) = \int_0^1 f(x,y)\,dx$$

$$= \int_0^1 (x+y)\,dx$$

$$= \int_0^1 x\,dx + y\int_0^1 dx$$

$$= \left| \frac{x^2}{2} \right|_0^1 + y\left| x \right|_0^1$$

$$= \frac{1}{2}\left(x^2 \right|_0^1 + y\,(1)$$

$$= 1/2 + y$$

$$f(x)_x = \int_0^1 x + y\,dy$$

$$= x\int_0^1 dy + \int_0^1 y\,dy$$

$$= x\left| y \right|_0^1 + \left| y^2/2 \right|_0^1$$

$$= x(1) + \frac{1}{2}\left| y^2 \right|_0^1$$

$$= x + 1/2\,(1)$$

$$f(x)_x = x + 1/2$$

Scanned with CamScanner

$$E(X) = \int_0^1 x\, f(x)\, dx$$

$$= \int_0^1 x\,(x + 1/2)\, dx$$

$$= \int_0^1 (x^2 + x/2)\, dx$$

$$= \left| \frac{x^3}{3} + \frac{x^2}{4} \right|_0^1$$

$$= 1/3 + 1/4$$

$$\Rightarrow 7/12$$

$$E(Y) = \int_0^1 y\, f(y)\, dy$$

$$= \int_0^1 y\,(y + 1/2)\, dy$$

$$= \int_0^1 (y^2 + y/2)\, dy$$

$$= \left| \frac{y^3}{3} + \frac{y^2}{4} \right|_0^1$$

$$= 1/3 + 1/4 \Rightarrow 7/12$$

$$E(X+Y) = E(X) + E(Y)$$

$$= 7/12 + 7/12$$

$$= 14/12$$

$$= 7/6$$

$$E(XY) = E(X)\,E(Y)$$

$$= (7/12)(7/12)$$

$$E(XY) = 49/144$$

Scanned with CamScanner

Q#03:-

Define lognormal dist and obtain its r^{th} moment about origion. if $\log x$ is normally distributed with $\mu = 1$, $\sigma^2 = 4$. Find $P(1/2 < X < 2)$

(Solution)

let "x" be a positive random variable and let y is a new random variable defined as $Y = \log x$. if Y has a normal-dist then x has a lognormal-dist. The p.d.f of lognormal-dist is given by

$$f(x) = f(x, \mu, \sigma) = \frac{1}{x\sqrt{2\pi}\,\sigma} e^{-1/2\sigma^2(\log x - \mu)^2} \quad 0 < x < \infty$$

r^{th} moment about origion

we know that the rth moment of lognormal-dist is

$$\mu_r' = e^{\mu r + r^2\sigma^2/2}$$

put $r = 1$

$$\mu_1' = e^{\mu + \sigma^2/2}$$

put $r = 2$

$$\mu_2' = e^{2\mu + 4\sigma^2/2}$$

$$= e^{2\mu + 2\sigma^2}$$

put $r = 3$

Scanned with CamScanner

$\mu_3' = e^{3\mu + 9\sigma^2/2}$

put $r = 4$

$\mu_4' = e^{4\mu + 16\sigma^2/2}$

$\mu_4' = e^{4\mu + 8\sigma^2/2}$

→ if $\log x$ is normaly-distributed with mean $= 1$, $\sigma^2 = 4$. Find $P(1/2 < x < 2)$

As we know that

$$f(x) = \frac{1}{x\sqrt{2\pi}\,\sigma} e^{-1/2\left(\log x - \mu/\sigma\right)^2}$$

$\mu = 1$, $\sigma = 2$

$$= \frac{1}{x\sqrt{2\pi}\cdot 2} e^{-1/2\left(\log x - 1/2\right)^2}$$

as $P(1/2 < x < 2)$

$$f(x) = \frac{1}{2\sqrt{2\pi}} \int_{1/2}^{2} \frac{1}{x} \cdot e^{-1/2\left(\log x - 1/2\right)^2} dx$$

$\because \log x = t$

$1/x\, dx = dt$

$dx = x\,dt$

$\text{if } x = 1/2 \Rightarrow t = \ln \frac{1}{2} \Rightarrow -\ln 2$

$x = 2 \Rightarrow t = \ln 2$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-\ln 2}^{\ln 2} \frac{1}{x} \cdot e^{-1/2\left(t - 1/2\right)^2} \cdot x\,dt$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-\ln 2}^{\ln 2} e^{-1/2\left(t - 1/2\right)^2} dt$$

Scanned with CamScanner

ibuted

$\frac{1}{2} < x < 2)$

$)^2$

dx

t

$= \frac{\ln 2 - 1}{2} \Rightarrow \ln 2$

$t = \ln 2$

$\therefore \frac{t-1}{2} = z$

$t = 1 + 2z$

$dt = 2dz$

$$= \frac{1}{\cancel{2}\sqrt{2\pi}} \int_{-\ln 2}^{\ln 2} e^{-z^2/2}\, \cancel{2} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\ln 2}^{\ln 2} e^{-1/2 z^2} dz$$

$\therefore \frac{\ln 2 - 1}{2} = z$

$-0.1534 = z$

$$= \frac{1}{\sqrt{2\pi}} \int_{-0.8467}^{-0.1534} e^{-1/2 z^2} dz$$

$\therefore \frac{-\ln 2 - 1}{2} = z$

$= -0.8467 = z$

$P((-0.8467) < z < (0.1534))$

$\phi(-0.1534) - \phi(-0.8467)$

$= 0.2996 - 0.0596$

$= 0.24$

Q # 5:-

Drive F-distribution

Solution:-

$$F(\chi_1^2) = \frac{1}{2^{\frac{n_1}{2}} \Gamma \frac{n_1}{2}} e^{-1/2(\chi_1^2)} (\chi_1^2)^{\frac{n_1}{2} - 1}$$

$$F(\chi_2^2) = \frac{1}{2^{\frac{n_2}{2}} \Gamma \frac{n_2}{2}}$$

Scanned with CamScanner

The joint dist of χ_1^2, χ_2^2

$$dF(\chi_1^2, \chi_2^2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} e^{-1/2(\chi_1^2+\chi_2^2)} (\chi_1^2)^{\frac{n_1}{2}-1} (\chi_2^2)^{\frac{n_2}{2}-1} d\chi_1^2 \, d\chi_2^2 \quad -(1)$$

→ Making the transformation

$$F = \frac{\chi_1^2/n_1}{\chi_2^2/n_2} \qquad \chi_2^2 = V$$

$$\frac{\chi_1^2}{n_1} = \frac{\chi_2^2}{n_2} F \qquad \frac{\partial \chi_1^2}{\partial F} = \frac{n_1}{n_2} V, \quad \frac{\partial \chi_1^2}{\partial \chi_2^2} = \frac{n_1}{n_2} F$$

$$\chi_1^2 = \frac{n_1}{n_2} VF \qquad \frac{\partial \chi_2^2}{\partial F} = 0, \quad \frac{\partial \chi_2^2}{\partial V} = 1$$

$$|J| = \begin{vmatrix} \frac{n_1}{n_2} V & \frac{n_1}{n_2} F \\ 0 & 1 \end{vmatrix}$$

$$J = \left| \frac{n_1}{n_2} V \right|$$

So, eq (1) is

$$dF(F, V) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} e^{-1/2\left(\frac{n_1}{n_2} VF + V\right)} \left(\frac{n_1}{n_2} VF\right)^{\frac{n_1}{2}-1} (V)^{\frac{n_2}{2}-1} \cdot \frac{n_1}{n_2} V \, dF \, dV$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} e^{-1/2 V\left(\frac{n_1}{n_2} F + 1\right)} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}-1+1} (V)^{\frac{n_2}{2}-1+1-1+\frac{n_1}{2}} F^{\frac{n_1}{2}-1+0}$$

$$= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \cdot F^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} dF \int_0^\infty e^{-1/2 V\left(\frac{n_1}{n_2} F + 1\right)} (V)^{\frac{n_1+n_2}{2}-1} dV$$

$$\because \frac{V}{2}\left(n_1+\frac{n_1}{n_2}F\right)=z$$

$$V=\frac{2z}{\left(1+\frac{n_1}{n_2}F\right)}$$

$$dV=\frac{2}{\left(1+\frac{n_1}{n_2}F\right)}dz$$

$V=0,\quad z=0$

$V=\infty,\quad z=\infty$

$$=\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}(F)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}}\Gamma\frac{n_1}{2}\Gamma\frac{n_2}{2}}dF\int_0^\infty e^{-z}\cdot\left(\frac{2z}{1+\frac{n_1}{n_2}F}\right)^{\frac{n_1+n_2}{2}-1}\cdot\left(\frac{2}{1+\frac{n_1}{n_2}F}\right)dz$$

$$=\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}(F)^{\frac{n_1}{2}-1}}{2^{\frac{n_1+n_2}{2}}\Gamma\frac{n_1}{2}\Gamma\frac{n_2}{2}}\left(\frac{2}{1+\frac{n_1}{n_2}F}\right)^{\frac{n_1+n_2}{2}}\int_0^\infty e^{-z}(z)^{\frac{n_1+n_2}{2}-1}dz$$

$$=\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}(F)^{\frac{n_1}{2}-1}(\cancel{2})^{\cancel{\frac{n_1+n_2}{2}}}}{\cancel{2^{\frac{n_1+n_2}{2}}}\Gamma\frac{n_1}{2}\Gamma\frac{n_2}{2}\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}\cdot\Gamma\frac{n_1+n_2}{2}$$

$$=\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}(F)^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2},\frac{n_2}{2}\right)\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}dF$$

$$d(F)=\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}(F)^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2},\frac{n_2}{2}\right)\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$

$0<F<\infty$

Scanned with CamScanner

Q:-06:-

Drive Characteristic function of cauchy-distribution

By defination

$$\phi(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} \cdot f(x) dx$$

$$= \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{\pi(1+x^2)} dx$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{(1+x^2)} dx \quad -(1)$$

→ By laplace-distribution

$$f(x) = \frac{1}{2} e^{-|x|}$$

$$\phi_x(t) = E(e^{itx})$$

$$= \int_{-\infty}^{\infty} e^{itx} \cdot f(x) dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} e^{itx} \cdot e^{-|x|} dx$$

$$\because e^{i} = \cos\theta + i\sin\theta$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} \cos(xt) + i\sin(xt) e^{-|x|} dx$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} \cos(xt) e^{-|x|} dx + \frac{1}{2} \int_{-\infty}^{\infty} i \sin x(t) e^{-|x|} dx$$

$$= \frac{\cancel{2}}{\cancel{2}} \int_{-\infty}^{\infty} \cos x(t) e^{-|x|} dx + 0$$

PUACP

Scanned with CamScanner

dion

By using $\int_{-\infty}^{\infty} e^{-mx} \cos nx = \frac{m^2}{m^2+n^2}$

$m = 1, \quad n = t$

$$\phi_x(t) = \frac{1}{(1+t^2)}$$

since $\phi(t)$ is absolutely integrable $(-\infty$ to $\infty)$, so by uniqness theorem

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx} \cdot \phi_x(t)\, dt$$

$$\frac{1}{\cancel{2}} e^{-|x|} = \frac{1}{\cancel{2}\pi}\int_{-\infty}^{\infty} e^{-itx} \cdot \frac{1}{1+t^2}\, dt$$

→ replace $-t \to t$

$$e^{-|x|} = \frac{1}{\pi}\int e^{itx} \cdot \frac{1}{1+t^2}\, dt$$

replace t by $\to x$

$$e^{-|t|} = \frac{1}{\pi}\int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{1+x^2}\, dx$$

→ replace x by x

$$e^{-|t|} = \frac{1}{\pi}\int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{1+x^2}\, dx$$

→ put in $\phi(t)$

$$\phi_x(t) = e^{-|t|}$$

$|x|$ dx

PUACP

Scanned with CamScanner

$$dv = \frac{2}{\left(1+\frac{n_1}{n_2}F\right)} \, d\chi$$

$F = 0 \quad , \quad \chi = 0$

$v = \infty \quad , \quad \chi = \infty$

d

$$d(F, v) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma\frac{n_1}{2} \Gamma\frac{n_2}{2}} \quad e$$

PUACP

Scanned with CamScanner

Probability

Past Paper

2017

Q:- 02:-

a) Find the mean deviation of the exponential distribution

(solution)

By defination

$$M.D = E|x-\mu|$$

$$= E|x - 1/\lambda| \quad \because \mu = 1/\lambda$$

$$= \int_0^\infty |x - 1/\lambda| \cdot f(x)dx$$

$$= \int_0^\infty |x - 1/\lambda| \cdot \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^\infty (x - 1/\lambda) e^{-\lambda x} dx$$

$$= \lambda \left[\int_0^{1/\lambda} -(x - 1/\lambda) e^{-\lambda x} dx + \int_{1/\lambda}^\infty (x - 1/\lambda) e^{-\lambda x} dx \right]$$

$$M.D = \lambda \left[-\int_0^{1/\lambda} x e^{-\lambda x} dx + \frac{1}{\lambda} \int_0^{1/\lambda} e^{-\lambda x} dx + \int_{1/\lambda}^\infty x e^{-\lambda x} dx - 1/\lambda \int_{1/\lambda}^\infty e^{-\lambda x} dx \right] \quad -(1)$$

Suppose

$$\int e^{-\lambda x} dx = \frac{e^{-\lambda x}}{-\lambda} \Rightarrow \frac{-1}{\lambda} e^{-\lambda x}$$

$$\int x e^{-\lambda x} dx = x \cdot \frac{e^{-\lambda x}}{-\lambda} - \int \frac{e^{-\lambda x}}{-\lambda} \cdot 1$$

Scanned with CamScanner

$$= -\frac{1}{\lambda} x e^{-\lambda x} - \frac{e^{-\lambda x}}{\lambda^2}$$

→ put this in eq(1)

$$M.D = \lambda \left\{ - \left| -\frac{1}{\lambda} x e^{-\lambda x} - \frac{e^{-\lambda x}}{\lambda^2} \right|_0^{1/\lambda} + \frac{1}{\lambda} \left| -\frac{1}{\lambda} e^{-\lambda x} \right|_0^{1/\lambda} \right.$$

$$+ \left| -\frac{1}{\lambda} x e^{-\lambda x} - \frac{e^{-\lambda x}}{\lambda^2} \right|_{1/\lambda}^{\infty} - \frac{1}{\lambda} \left| -\frac{1}{\lambda} e^{-\lambda x} \right|_{1/\lambda}^{\infty}$$

$$= \cancel{\lambda} \cdot \frac{1}{\cancel{\lambda}} \left[\left| x e^{-\lambda x} + \frac{e^{-\lambda x}}{\lambda} \right|_0^{1/\lambda} - \frac{1}{\lambda} \left| e^{-\lambda x} \right|_0^{1/\lambda} - \left| x e^{-\lambda x} + \frac{e^{-\lambda x}}{\lambda} \right|_{1/\lambda}^{\infty} \right.$$

$$+ 1/\lambda \left| e^{-\lambda x} \right|_{1/\lambda}^{\infty}$$

$$= \frac{1}{\lambda} e^{-1} + \frac{1}{\lambda} e^{-1} - (1/\lambda\, e^{0}) - \frac{1}{\lambda} (e^{-1} - e^{0}) -$$

$$\left\{ x e^{-\infty} + 1/\lambda\, e^{-\infty} - (1/\lambda\, e^{-1} + \frac{1}{\lambda} e^{-1}) + 1/\lambda (e^{-\infty} - e^{-1}) \right.$$

$$= \frac{1}{\lambda} e^{-1} + \frac{1}{\lambda} e^{-1} - \cancel{\frac{1}{\lambda}} - \cancel{\frac{1}{\lambda} e^{-1}} + \cancel{\frac{1}{\lambda}} + \cancel{\frac{1}{\lambda} e^{-1}} +$$

$$\cancel{1/\lambda\, e^{-1}} - \cancel{1/\lambda\, e^{-1}}$$

$$M.D = \frac{2}{\lambda} e^{-1}$$

d)- Show that

$$t^2(n) = F(1, n)$$

(solution)

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} \cdot F^{\frac{n_2}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\left(1+\frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}$$

Put $n_1 = 1, n_2 = n,$

$$F = t^2 \qquad t=0, F=0$$

$$dF = 2t\,dt \qquad t=\infty, F=\infty$$

$$f(t) = \frac{\left(\frac{1}{n}\right)^{1/2} \left(t^2\right)^{n/2-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right)\left(1+\frac{1}{n}t^2\right)^{\frac{1+n}{2}}} 2t\,dt$$

$$= \frac{2\left(t^2\right)^{-1/2} t}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{1}{n}t^2\right)^{\frac{1+n}{2}}} dt \qquad 0<t<\infty$$

$$= \frac{2}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{1+n}{2}}} dt \qquad 0<t<\infty$$

it is an even function

$$f(t) = \frac{1}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{1+n}{2}}} dt$$

$$f(t) = \frac{1}{\sqrt{n}\,\beta\left(\frac{1}{2}, \frac{n}{2}\right)\left(1+\frac{t^2}{n}\right)^{\frac{1+n}{2}}}$$

$$0<t<\infty$$

Scanned with CamScanner

e) Find moment generating function of t-distribution

-(sol)-

The moment generating function of t-distribution does not exist

F) if $X \sim N(\mu, \sigma^2)$ then find the mean and variance of $Y = aX + b$

-(sol)-

$$Y = aX + b$$

Mean properties

$$\mu_Y = a\mu_X + b$$

$$\mu_Y = a\mu + b$$

$$\sigma_X^2 = \sigma$$

$$V(Y) = a^2 V(X)$$

$$\sigma_Y^2 = a^2 \sigma^2$$

g) Define distribution function and its properties

-(sol)-

if X is a random variable its distribution function is a function

$F_X : R \to [0, 1]$ such that

$$F_X(x) = P(X \leq x) \quad \forall x \in R$$

where $P(X \leq x)$ is the probability that X is less than or equal to x.

Scanned with CamScanner

Properties:

Increasing → $F_x(x)$ is increasing

$F_x(x_1) \leq F_x(x_2)$ if $x_1 < x_2$

Right continuoution → $F_x(x)$ is Right continoution

$$\lim_{t \to x} F_x(t) = F_x(x)$$

for any $x \in R$

limit at minus infinity: $F_x(x)$ satisfies

$$\lim_{x \to 0} F(x) = 0$$

limit at plus infinity: $F_x(x)$ satisfies

$$\lim_{x \to \infty} F(x) = 1$$

Q:03

(a) if "x" follows two parameter Gamma distribution with density function

$$f(x) = \frac{x^{\alpha-1} e^{-(x/\beta)}}{\Gamma\alpha \, \beta^{\alpha}} \; ; 0<x<\infty, \alpha>0, \beta>0$$

Drive its moments generating function and use it to find four comulants and γ_1 and γ_2

(solution)

$$f(x) = \frac{x^{\alpha-1} \, e^{-(x/\beta)}}{\Gamma\alpha \, \beta^{\alpha}} \quad 0<x<\infty$$

By defination

$$M_0(t) = E(e^{tx})$$

$$= \int_0^{\infty} e^{tx} \cdot f(x) \, dx$$

$$= \int_0^{\infty} e^{tx} \cdot \frac{x^{\alpha-1} e^{-(x/\beta)}}{\Gamma\alpha \, \beta^{\alpha}} \, dx$$

$$= \frac{1}{\Gamma\alpha \, \beta^{\alpha}} \int_0^{\infty} x^{\alpha-1} e^{-x(1/\beta - t)} \, dx$$

$$\because x\left(\frac{1-\beta t}{\beta}\right) = y$$

$$= \frac{1}{\Gamma\alpha \, \beta^{\alpha}} \int_0^{\infty} \left(\frac{\beta}{1-\beta t}\right)^{\alpha-1} y^{\alpha-1} \cdot e^{-y} \cdot \frac{\beta}{(1-\beta t)} dy \quad \because x = \left(\frac{\beta}{1-\beta t}\right) y$$

$$\because dx = \frac{\beta}{(1-\beta t)} dy$$

$$= \frac{\left(\beta/(1-\beta t)\right)^{\alpha-\cancel{1}+\cancel{1}}}{\Gamma\alpha \, \beta^{\alpha}} \int_0^{\infty} e^{-y} \cdot y^{\alpha-1} dy$$

Scanned with CamScanner

$$= \frac{\cancel{(\beta)^{\alpha}} / (1-\beta t)^{\alpha}}{\cancel{\beta^{\alpha}}\ \cancel{\Gamma\alpha}} \cdot \cancel{\Gamma\alpha}$$

$$M_0(t) = (1-\beta t)^{-\alpha}$$

→ Taking ln on b/s

$$\ln M_0(t) = \ln(1-\beta t)^{-\alpha}$$

$$= -\alpha \ln(1-\beta t)$$

$$= \alpha(-\ln(1-\beta t))$$

$$\because \ln(1-x) = x + x^2/2 + x^3/3 + \cdots$$

$$= \alpha\left\{(\beta t) + \frac{(\beta t)^2}{2} + \frac{(\beta t)^3}{3} + \frac{(\beta t)^4}{4} + \cdots\right\}$$

$$= \alpha\beta t + \alpha\beta^2 \frac{t^2}{2!} + 2\alpha\beta^3 \frac{t^3}{3!} + 6\alpha\beta^4 \frac{t^4}{4!} + \cdots$$

→ By comparing cofficients $t^r/r!$

$K_1 = \alpha\beta$

$K_2 = \alpha\beta^2$

$K_3 = 2\alpha\beta^3$

$K_4 = 6\alpha\beta^4$

$$\gamma_1 = (K_3)^2 / (K_2)^{3/2}$$

$$= \frac{(2\alpha\beta^3)^2}{(\alpha\beta^2)^{3/2}}$$

$$= \frac{2\alpha\cancel{\beta^3}}{\alpha^{3/2} \cdot \cancel{\beta^3}}$$

$$= 2\alpha^{1-3/2}$$

$$= 2/\sqrt{\alpha}$$

Scanned with CamScanner

$$\gamma_2 = k_4 / (k_2)^2$$

$$= \frac{6\alpha\beta^4}{(\alpha\beta^2)^2}$$

$$= \frac{6\alpha\beta^4}{\alpha^2\beta^4}$$

$$\gamma_2 = \frac{6}{\alpha}$$

b) Show that marginal density of "x" in Bivariate normal dist follows uni-variate Normal-dist

-(sol)-

The marginal density function of variable x is

$$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[\frac{-1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(y-\mu_2/\sigma_2\right)^2\right\}\right]$$

put $\frac{y-\mu_2}{\sigma_2} = v$

$$y = \mu_2 + v\sigma_2$$

$$dy = 0 + \sigma_2 dv$$

$$= \frac{1}{2\pi\sigma_1\cancel{\sigma_2}\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[\frac{-1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)v + v^2\right\}\right] \cdot \cancel{\sigma_2}\, dv$$

Add and subtract $\rho^2 \left(\frac{x-\mu_1}{\sigma_1}\right)^2$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(\frac{-1}{2(1-\rho^2)}\right)\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right) v + (v)^2 + \rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \rho^2\left(\frac{x-\mu_1}{\sigma_1}\right)^2\right\} dv$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left\{\frac{-1}{2(1-\rho^2)}\right\}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2(1-\rho^2) + \left(v-\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right)^2\right\} dv$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[\frac{-1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\frac{\left(v-\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\right)^2}{1-\rho^2}\right]$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[\frac{-1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}\left(\frac{v-\rho\left(\frac{x-\mu_1}{\sigma_1}\right)}{\sqrt{1-\rho^2}}\right)^2\right]$$

$$\because \frac{v-\rho(x-\mu_1/\sigma_1)}{\sqrt{1-\rho^2}} = w$$

$$v-\rho(x-\mu_1/\sigma_1) = w\sqrt{1-\rho^2}$$

$$dv = dw\sqrt{1-\rho^2}$$

$$g(x) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[\frac{-1}{2}\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - \frac{1}{2}w^2\right]\cancel{\sqrt{1-\rho^2}}\, dw$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot e^{-1/2 w^2}\, dw$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{2\pi}\,\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot e^{-1/2 w^2}\, dw$$

Scanned with CamScanner

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 w^2} \cdot dw$$

$$\because \frac{w^2}{2} = t$$

$$w^2 = 2t$$

$$\cancel{2} w dw = \cancel{2} dt$$

$$dw = \frac{dt}{\sqrt{2t}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot \frac{\cancel{2}}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} \cdot \frac{dt}{\sqrt{2t}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-t} \cdot t^{-1/2} dt$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-t} \cdot t^{1/2-1} dt$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2} \cdot \frac{1}{\cancel{\sqrt{\pi}}} \cdot \cancel{\sqrt{\pi}}$$

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-1/2(x-\mu_1/\sigma_1)^2}$$

c) State and drive the probability density function of t-dist

(sol)

By defination t-dist is

$$t = \frac{x}{\chi\sqrt{n}}$$

where x and χ (chi) are independent, x is standard normal with mean 0 and variance 1 and

Scanned with CamScanner

χ (chi) is distributed with "n" degrees of freedom

(proof)

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-1/2 x^2} \, dx$$

$$f(\chi) = \frac{2}{2^{n/2} \sqrt{\frac{n}{2}}} e^{-1/2 \chi^2} \chi^{n-1} \qquad 0 < \chi < \infty$$

The joint distribution of x and χ

$$dF(x, \chi) = \frac{2}{\sqrt{2\pi}\, 2^{n/2} \sqrt{\frac{n}{2}}} e^{-1/2 (x^2 + \chi^2)} \chi^{n-1} \, dx \, d\chi$$

let

$x = r\cos\theta \quad \partial x/\partial r = +\cos\theta, \ \partial x/\partial\theta = -r\sin\theta$

$\chi = r\sin\theta, \quad \partial \chi/\partial r = \sin\theta, \ \partial \chi/\partial\theta = r\cos\theta$

$$|J| = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= \left| r\cos^2\theta + r\sin^2\theta \right|$$

$$= r\left(\cos^2\theta + \sin^2\theta\right)$$

$$|J| = r$$

$$= \frac{2}{\sqrt{2\pi}\, 2^{n/2} \sqrt{\frac{n}{2}}} e^{-1/2 r^2} (r\cos\theta)^{n-1} \, r \, dr \, d\theta$$

$$= \frac{2}{\sqrt{2\pi}\, 2^{n/2} \sqrt{\frac{n}{2}}} (\cos\theta)^{n-1} d\theta \int_0^\infty e^{-1/2 r^2} \cdot r^n \, dr$$

$A \cap B$

$\because \frac{r^2}{2} = z$

$r^2 = 2z$

$\cancel{2}r\,dr = \cancel{2}\,dz$

$dr = \frac{dz}{\sqrt{2z}}$

$$= \frac{2}{\sqrt{2\pi}\, 2^{n/2} \Gamma\frac{n}{2}} (\cos\theta)^{n-1} d\theta \int_0^\infty e^{-z} \cdot \frac{dz}{\sqrt{2z}} \cdot (2z)^{n/2}$$

$$= \frac{2 \cdot 2^{n/2 - 1/2} (\cos\theta)^{n-1} d\theta}{\sqrt{2\pi}\, 2^{n/2} \Gamma\frac{n}{2}} \int_0^\infty e^{-z} \cdot (z)^{n/2 - 1/2}\, dz$$

$$= \frac{2^{1 + \cancel{n/2} - 1/2 - 1/2 - \cancel{n/2}}}{\sqrt{\pi} \cdot \Gamma\frac{n}{2}} (\cos\theta)^{n-1} d\theta \int_0^\infty e^{-z} \cdot (z)^{n/2 + 1/2 - 1} dz$$

$$= \frac{1}{\Gamma\frac{1}{2} \cdot \Gamma\frac{n}{2}} \cdot (\cos\theta)^{n-1} \cdot d\theta\, \Gamma\frac{n+1}{2}$$

$$= \frac{(\cos\theta)^{n-1}}{\frac{\Gamma\frac{1}{2}\Gamma\frac{n}{2}}{\Gamma\frac{n+1}{2}}} d\theta$$

$$dF(\theta) = \frac{(\cos\theta)^{n-1}}{\beta\left(\frac{1}{2}, \frac{n}{2}\right)} d\theta$$

As

$$t = x \Big/ x / \sqrt{n}$$

$$= \frac{\cancel{r}\sin\theta \sqrt{n}}{\cancel{r}\cos\theta}$$

$$t = \tan\theta \sqrt{n}$$

Scanned with CamScanner

$$dt = \sqrt{n}\sec^2\theta\, d\theta$$

$$t^2 = (n)\tan^2\theta$$

$$t^2/n = \tan^2\theta$$

$$\frac{t^2}{n} = \sec^2\theta - 1$$

$$1+\frac{t^2}{n} = \sec^2\theta$$

$$1/(1+t^2/n) = \cos^2\theta$$

$$d\theta = \frac{dt}{\sqrt{n}\sec^2\theta}$$

$$d\theta = \frac{dt}{\sqrt{n}(1+t^2/n)}$$

Hence the distribution of t

$$= \frac{1}{\beta(\frac{1}{2},\frac{n}{2})} \cdot \left(\frac{1}{(1+t^2/n)}\right)^{n-1} \cdot \frac{dt}{\sqrt{n}(1+t^2/n)}$$

Q#05:-

a) if x has an Exponential distribution given by

$$f(x) = \frac{1}{2} e^{-(x/2)} \qquad 0 \le x \le \infty$$

what are the mean, variance and m.g.f of x? Also calculate $p(x<3)$ and $p(x>5 \mid x>2)$

The Exponential dist is given

$$f(x) = \frac{1}{2} e^{-(x/2)}$$

By defination

$$M_o(t) = E(e^{tx})$$

$$M_o(t) = \int_0^\infty e^{tx} \cdot f(x) dx$$

$$= \frac{1}{2}\int_0^\infty e^{tx} \cdot e^{-x/2} dx$$

$$= \frac{1}{2}\int_0^\infty e^{-x(1/2 - t)} dx$$

put $x(1/2 - t) = z$

$$x = \frac{2z}{1-2t}$$

$$dx = \frac{2}{1-2t} dz$$

$$= \frac{1}{2}\int_0^\infty e^{-z} \cdot \frac{2}{1-2t} dz$$

$$= \frac{1}{1-2t}\int_0^\infty e^{-z} \cdot z^{1-1} dz$$

$$= \frac{1}{1-2t} \Gamma 1$$

$$M_o(t) = (1-2t)^{-1}$$

→ Now diff w.r.t "t"

$$\frac{dM_o(t)}{dt} = -1(1-2t)^{-2} \cdot (-2)$$

$$= 2(1-2t)^{-2}$$

$$t = 0$$

$$= 2(1-2(0))^{-2}$$

Scanned with CamScanner

$= 2(1-0)^{-2}$

Mean $= 2$

Now. diff w.r.t "t"

$= -4(1-2t)^{-3} \cdot (-2)$

$= 8(1-2t)^{-3}$

put $t = 0$

$= 8(1-2(0))^{3}$

$E(x^2) = 8$

$VAR(x) = E(x^2) - (E(x))^2$

$= 8 - (2)^2$

$= 8 - 4$

$VAR(x) = 4$

$P(x < 3)$

0 to 3

$$= \int_0^3 q(x)\, dx$$

$$= \int_0^3 \frac{1}{2} e^{-x/2}\, dx$$

$$= \frac{1}{2} \int_0^3 e^{-x/2}\, dx$$

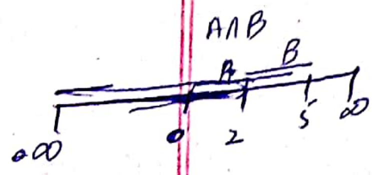
$$= \frac{1}{\cancel{2}} \left| \frac{e^{-x/2}}{-1/\cancel{2}} \right|_0^3$$

$$= -\left(e^{-3/2} - e^{-0}\right)$$

$$= -\left(e^{-3/2} - 1\right)$$

$$= 1 - e^{-3/2}$$

Scanned with CamScanner



$P(x>5 \mid x>2)$

$P(2<x<5) \to P(2<x<\infty)$

$$f(x) = \int_2^5 \frac{1}{2} e^{-x/2}\, dx$$

$$= \frac{1}{2}\int_2^5 e^{-x/2}\, dx$$

$$= \frac{1}{\cancel{2}} \left| \frac{e^{-x/2}}{-1/\cancel{2}} \right|_2^5$$

$$= -\left| e^{-5/2} - e^{-2/2} \right|$$

$$= -\left(e^{-5/2} - e^{-1}\right)$$

$$= \left(e^{-1} - e^{-5/2}\right)$$

$$= 1/e - 1/e^{5/2}$$

Now $P(2<x<\infty)$

$$= \frac{1}{2}\int_2^\infty e^{-x/2}\, dx$$

$$= \frac{1}{\cancel{2}} \left| \frac{e^{-x/2}}{-1/\cancel{2}} \right|_2^\infty$$

$$= -1\left(e^{-\infty/2} - e^{-2/2}\right)$$

$$= -1\left(0 - e^{-1}\right)$$

$$= \left(e^{-1}\right)$$

$$f(x) = e^{-1}$$

b)- Five observations x_1, x_2, x_3, x_4, x_5 are drawn at random from the distribution

$$f(x) = 5e^{-5x} \qquad 0 \le x \le \infty$$

What is the distribution of (i)

i)- The smallest observation

ii)- The largest observation

—(sol)—

$$f(x) = 5e^{-5x}$$

$$F(x) = 5\int_0^x e^{-5x}\,dx$$

$$= 5\left|\frac{e^{-5x}}{-5}\right|_0^x$$

$$= -(e^{-5x} - e^0)$$

$$= (e^0 - e^{-5x})$$

$$F(x) = 1 - e^{-5x}$$

The r^{th} order statistic is

$$g(y_r) = \frac{n!}{(r-1)!\,(n-r)!}\left[F(y_r)\right]^{r-1} f(y_r)\left[1-F(y_r)\right]^{n-r}$$

i- The Smallest observation

put $r=1$, $n=5$

$$= \frac{5!}{(1-1)!\,(5-1)!}\left[F(y_1)\right]^{1-1} f(y_1)\left[1-F(y_1)\right]^{5-1}$$

$$= \frac{5\times\cancel{4!}}{\cancel{4!}} f(y_1)\left[1-F(y_1)\right]^4$$

$$= 5\left(5e^{-5x}\right)\left[1-(1-e^{-5x})\right]^4$$

Scanned with CamScanner

Questi

I

II

$$= 25\left(e^{-5x}\right)\left(\cancel{1}-\cancel{1}+e^{-5x}\right)^4$$

$$= 25\left(e^{-5x-20x}\right)$$

$$= 25\left(e^{-25x}\right)$$

ii)- The largest observation

$r = n$, $n = 5$

$$= \frac{5!}{(5-1)!\,(5-5)!}\left[F(y_5)\right]^{5-1} f(y_5)\left[1-F(y_5)\right]^{5-5}$$

$$= \frac{5 \times \cancel{4!}}{\cancel{4!}}\left[F(y_5)\right]^{4} f(y_5)$$

$$= 5\left(5e^{-5x}\right)\left[1-e^{-5x}\right]^4$$